

LAMPIRAN SOAL (KERJAKAN MENGGUNAKAN TAHAPAN BASIS PATH)

Soal 1 (15 Poin)

TES1.PAS
<pre>Program Coba; Var a , b : integer; Begin a := 5; {1} b := 10; {2} writeln; {3} end.</pre>

Soal 2 (20 Poin)

TES2C.PAS
<pre>Program Coba; Var a , b : integer; Begin a := 5; {1} b := 8; if a > b then {2} a := 2 {3} else b := a; end.</pre>

Soal 3 (15 Poin)

TES2D.PAS

```
Program Coba;  
Var a , b : integer;  
Begin  
    A := 5;           {1}  
    If a > 3 then     {2}  
        a := 2;  
        writeln(a);  
end.
```

Soal 4 (20 Poin)

TES3.PAS
<pre>Program Coba; var a : integer; begin a := 5; while a < 10 do begin a := a + 1; end; a := a + 10; writeln; end.</pre>

Soal 5 (15 Poin)

TES4.PAS
<pre>Program Coba; Var a : integer; Begin for a := 5 to 10 do begin writeln(a); end; writeln; end.</pre>

Soal 6 (20 Poin)

TES5.PAS
<pre>Program Coba; Var a : integer; Begin a := 5; repeat a := a + 1; until a > 10; a := a + 20; writeln; end.</pre>

Soal 7 (40 Poin)

TES7.PAS

```
Program Coba;
var a , b, c : integer;
begin
    a := 5;
    b := 8;
    c := 4;
    if (a > b) or (a > c) then
    begin
        a := 2;
    end
    else
    begin
        if b = a then
            b := a
        else
            b := c;
        end;
    readln;;
end.
```

Soal 8 (30 Poin)

TES8.PAS

```
Program Coba;
var a : integer;
begin
    a := 5;
    while a < 10 do
    begin
        if a > 5 then
            writeln(a);
            a := a + 1;
        end;
        a := a + 10;
        writeln;
    end.
```

Soal 9 (30 Poin)

TES9.PAS

<pre>Program Coba; var a, b : integer; begin readln (b); for a := 1 to 10 do begin if a > b then writeln(a) else writeln(b); end; writeln(a, b); end.</pre>

Soal 10 (40 Poin)

TES12.PAS

<pre>Program Coba; var a : integer; begin a := 5; if a > 4 then begin while a < 10 do begin if a > 5 then writeln(a); a := a + 1; end; end else a := a + 10; writeln; end.</pre>

Soal 11 (50 Poin)

TES27.PAS

```
Program Coba;
var a , b, c : integer;
begin
    a := 5;
    b := 8;
    c := 4;
    if a > b then
    begin
        a := 2;
    end
    else
    begin
        case b of
            1 : b := a;
            2 : b := a + 1;
            else
                b := c;
            end;
        end;
        writeln(a,b);
    end.
```

Soal 12 (50 Poin)

ALGO19.PAS

```
Program BinomialCoefficients;
Var
    i, a, b : integer;
    m, n     : integer;
begin
    readln( m, n );
    a := 1;
    if 2 * m > n then
        b := n - m
    else
        b := m;
    i := 0;
    while i <= b do
    begin
        a := (n - i ) * a + (i +1);
        i := i + 1;
    end;
    writeln( a );
end.
```

Soal 13 (80 Poin)

```
ALGOL06.PAS
Program ComplexNumberToARealPower;
Var
  x, y, w, A, B : real;
  theta, phi, r : real;
begin
  readln( x, y, w, A, B );
  if x > 0 then
  begin
    theta := 0.0;
    phi := arctan(y/x)+theta;
    r := sqrt( x*x+y*y );
    r := exp( w * ln(r));
    a := r * cos( w * phi);
    b := r * sin( w * phi);
  end;

  if (x < 0) and (y >= 0) then
  begin
    theta := 3.1;
    phi := arctan(y/x)+theta;

    r := sqrt( x*x+y*y );
    r := exp( w * ln(r));
    a := r * cos( w * phi);
    b := r * sin( w * phi);
  end;
  if (x<0) and (y<0) then
  begin
    theta := 3.1;
    phi := arctan(y/x)+theta;

    r := sqrt( x*x+y*y );
    r := exp( w * ln(r));
    a := r * cos( w * phi);
    b := r * sin( w * phi);
  end;
  if (x=0) and (y= 0) then
  begin
    a := 0.0;
    b := 0.0;
  end;

  if (x = 0) and (y < 0) then
  begin
    phi := 1.5;
    r := sqrt( x*x+y*y );
    r := exp( w * ln(r));
    a := r * cos( w * phi);
    b := r * sin( w * phi);
  end;

  if (x = 0) and (y > 0) then
  begin
    phi := -1.5707963;
    r := sqrt( x*x+y*y );
    r := exp( w * ln(r));
    a := r * cos( w * phi);
    b := r * sin( w * phi);
  end;
  writeln(a,b)
end.
```